Siyuan (Max) Meng

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Education

- **University of Massachusetts Amherst**, USA 2023.09 – 2026.08 (Expected) Ph.D. in Civil Engineering (Transportation Engineering), Fully Funded GPA: 3.97/4.0 Advisor: Dr. Chengbo Ai
- **Chang'an University**, China 2020.09 – 2023.07 M.S. in Transportation Engineering GPA: 3.74/4.0 Advisor: Dr. Qiang Bai
- **Chang'an University**, China 2016.09 – 2020.07 B.S. in Roadway, Bridge, and River-Crossing Engineering GPA: 3.76/4.0

Research Vision

My research centers on **Infrastructure-Centric World Models (I-WM)**—a framework that leverages the unique "temporal depth" of fixed roadside sensors to build rich, evolving representations of traffic environments. This vision unifies my doctoral work on quality-aware multi-LiDAR perception (background modeling → detection → tracking → safety analysis) with emerging directions in V2X cooperative perception and spatial intelligence. A second research thrust extends multi-objective optimization methods to electric vehicle charging infrastructure and transportation asset management, addressing critical needs in sustainable and intelligent transportation systems. See: arXiv:2604.17651.

Research Experience

- **Graduate Research Assistant**, UMass Amherst 2023.09 – Present
  Infrastructure-Centric Multi-LiDAR Perception Pipeline:
  - Developed FRGB3D, a fast reliability-weighted Gaussian background model for roadside LiDAR that propagates sensor-level uncertainty into foreground extraction. Published in *ASCE JCCE*.
  - Designed MulDet3D, a multi-objective optimization-based unsupervised 3D detection framework that jointly calibrates and detects objects across multiple roadside LiDARs without manual annotation. Under review at *JTE Part A*.
  - Built MulTrack3D, a reliability-enhanced self-correcting multi-point tracking system for roadside LiDAR networks with quality-aware data association. Under review at *JTE Part A*.
  - Developed a quality-aware vehicle conflict detection method with spatial precision enhancement for traffic safety analysis from roadside multi-LiDAR systems.

  Transportation System Optimization:
  - Formulated a robust MPC framework for system-optimal EV charging scheduling under demand uncertainty with mixed autonomous/human fleet routing.
  - Developing a multi-objective optimization framework for ultra-fast EV charging infrastructure planning under power grid constraints.
- **Graduate Research Assistant**, Chang'an University 2020.09 – 2023.07
  Developed multi-objective optimization methods for network-level pavement maintenance planning, including segment grouping strategies and user cost modeling. Published in *ASCE JIS* and *TRR*.
- **Teaching Assistant**, UMass Amherst 2024
  Course: Programming for Civil Engineers (Python-based).

Publications

First-Author Journal Articles

1. **Meng, S.**, Parasha, P., Yang, Y., & Ai, C. (2026). FRGB3D: Fast Reliability-Weighted Gaussian Background Modeling for Roadside LiDAR Traffic Monitoring. *Journal of Computing in Civil Engineering.* **(Accepted)**
2. **Meng, S.**, Bai, Q., & Shao, Y. (2026). A multi-objective optimization method for integrated road asset management considering traffic dynamics and environmental impacts. *Transportation Research Record.* **(Accepted)**
3. **Meng, S.**, Bai, Q., Chen, L., & Hu, A. (2023). Multiobjective optimization method for pavement segment grouping in multiyear network-level planning of maintenance and rehabilitation. *Journal of Infrastructure Systems*, 29(1), 04022047.

First-Author Papers Under Review

1. **Meng, S.**, Zhang, Y., Raucci, D., & Ai, C. MulDet3D: Multi-Objective Optimization Based Unsupervised Object Detection for Multiple Roadside LiDARs. *Journal of Transportation Engineering Part A: Systems.* (Under Review after Revision)
2. **Meng, S.**, Cheng, R., & Ai, C. MulTrack3D: Reliability-Enhanced Self-Correcting Multi-Point Tracking for Multiple Roadside LiDARs. *Journal of Transportation Engineering Part A: Systems.* (Under Review)
3. **Meng, S.** & Ai, C. A Robust MPC Framework for System-Optimal EV Charging Scheduling under Demand Uncertainty. *Transportation Research Part C: Emerging Technologies.* (Under Review)
4. **Meng, S.**, Yang, Y., & Ai, C. Quality-Aware Vehicle Conflict Detection with Spatial Precision Enhancement for Roadside Multi-LiDAR Systems. Target: *Accident Analysis & Prevention.* (Submitting)

First-Author Preprint

1. **Meng, S.** & Ai, C. (2026). Infrastructure-Centric World Models: Bridging Temporal Depth and Spatial Breadth for Roadside Perception. *arXiv preprint arXiv:2604.17651.* [Link]

First-Author Conference Paper

1. **Meng, S.**, Shao, Y., Cao, R., & Wu, S. (2022). Multi-objective optimization of network-level pavement maintenance decisions considering user costs [in Chinese]. In *Proc. World Transport Convention (WTC 2022)*, pp. 693–701. doi.

Co-Authored Publications

1. Hu, A., Bai, Q., Chen, L., **Meng, S.**, Li, Q., & Xu, Z. (2022). A review on empirical methods of pavement performance modeling. *Construction and Building Materials*, 342, 127968. (Cited 75+)
2. Cao, R., **Meng, S.**, Feng, S., Dong, S., & Bai, Q. (2024). Network-level pavement maintenance and rehabilitation planning considering uncertainties. *International Journal of Pavement Engineering*, 25(1), 2340001.
3. Bai, Q., Wu, S., Cao, R., **Meng, S.**, & Xu, Z. (2022). Review of optimization methods for civil airport passenger boarding processes. *Journal of Traffic and Transportation Engineering* [in Chinese], 22(04), 68–88.
4. Bai, Q., Shao, Y., **Meng, S.**, Wang, Y., Chen, X., & Feng, H. (2022). Video-based detection method for illegal pick-up vehicles at airport departure levels. *Journal of Chang'an University (Natural Science Edition)* [in Chinese], 42(04), 73–86.
5. Fang, Z., **Meng, S.** (2025). Estimating travel mode choices based on multinomial logit and mixed logit models. *Proc. SPIE 13575, STCE 2024*, 135753V.
6. Li, C., **Meng, S.** (2025). Analysis of expressway maintenance decision based on game theory. *Proc. SPIE 13575, STCE 2024*, 135751T.

Selected Presentations

- **Meng, S.** (2026). FRGB3D: Fast Reliability-Weighted Gaussian Background Modeling for Roadside LiDAR Traffic Monitoring. *TRB 2026 Annual Meeting*, Washington D.C.
- **Meng, S.** (2026). Quality-Aware Vehicle Conflict Detection for Roadside Multi-LiDAR Systems. *Safety Mobility Conference 2026*, Seattle, WA.
- **Meng, S.** (2025). Multi-objective Optimization for Integrated Roadway Asset Maintenance. *TRB 2025 Annual Meeting*, Washington D.C.

Patents & Software

- Bai, Q., Shao, Y., **Meng, S.**, et al. (2025). Deep learning-based automatic detection of illegal vehicles at airport departure levels. Granted Patent No. 202111642183.3 [in Chinese].
- **Meng, S.**, et al. (2023). Airport curbside illegal vehicle monitoring system. Software Copyright [in Chinese].

Honors & Awards

- Outstanding Graduate Student, Chang'an University (2023)
- First Prize Scholarship, Chang'an University (2021)
- Second Prize, English Speech Contest, Northwest China (2021)
- Excellent Undergraduate Thesis, Chang'an University (2020)

Technical Skills

Programming: Python, C++, MATLAB, LaTeX **ML/CV:** PyTorch, TensorFlow, OpenCV, Open3D **Simulation:** CARLA, SUMO, RoadRunner, ROS2 **Other:** GIS, SLAM, optimization modeling, LiDAR point cloud processing

Professional Service

Reviewer for peer-reviewed journals in transportation and pavement engineering.